Exp-12 End-End Communication at Transport Layer.

ei) Aim:-

To Implement echo client server using TCP/UDP sockets.

Server:-

```
import socket

s = socket.socket()
s.bind(("127.0.0.1", 5000))
s.listen(1)

c, addr = s.accept()
msg = c.recv(1024).decode()
print("client: ", msg)
c.send(msg.encode())
c.close()
s.close()
```

client:-

```
import socket

s = socket.socket()
s.connect(("127.0.0.1", 5000))

msg = input("Enter message: ")
s.sent(msg.encode())
print("Server: ", s.recv(1024).decode())
s.close()
```

Scanned with OKEN Scanner

input

Enter message: Hello Server

output

client connected

Received : Hello server

client echo from server : Hello Server

Result:

Thus to implement echo client server using TCP/UDP sockets are implemented successfully.

Scanned with OKEN Scanner

b) Aim:

To implement chat client server using TCP/UDP sockets.

Server:

```
import socket
s.bind(("127.0.0.1", 5001))
s.listen(1)
c, addr = s.accept()
print("Client: ", c.recv(1024).decode())
c.send(input("Server:").encode())
c.close()
s.close()
```

Client:

```
import socket
s = socket.socket()
s.connect(("127.0.0.1", 5001))
s.send(input("client:").encode())
print("Server:", s.recv(1024).decode())
s.close()
```

Scanned with OKEN Scanner

Input:-

client : Hello

Server : Hi

client : Bye

Output:-

client : Hello

Server : Hi

Result:-

Thus to implement chat client server using TCP/UDP sockets was completed successfully.